# Lab 4 — Predict Through Time

**Friday, November 6 (~20 min)**

Last updated: October 12, 2026.

In class Friday, November 6 (same day as Lecture 32 / Quiz 9). Quiz 9 is Week 10 products / cost — **not** this lab.

**Credit:** Lab points are for effort in class, not for a Canvas file upload. Preview the one-pager; write on the hard copy handed out in class.

## Learning objectives

1. Implement one step of $x_{t+1} = Ax_t$.
2. Iterate several steps and hand-check the first ones.
3. Describe a long-run pattern you observe (you will explain it in Lab 5).

## Math connections

Linear dynamical systems (Lectures 24–26).

## Key syntax (this lab)

| Task | Octave | Python |
|---|---|---|
| One step | `x = A*x` | `x = A @ x` |
| Loop | `for t = 1:20 … end` | `for t in range(20):` |
| Plot component | `plot(0:N, X(1,:))` | `plt.plot(range(N+1), X[0])` |

## Ops strip

On-/off-campus counts $x_t = \begin{bmatrix} c_t \\ o_t \end{bmatrix}$. Each week:

$$A = \begin{bmatrix} 0.7 & 0.2 \\ 0.3 & 0.8 \end{bmatrix}.$$

**C4.0.** Enter $A$ and $x_0 = (1000, 0)$. Print $A * x_0$ / `A @ x0` (that is $x_1$).

**Expected:** $x_1 = (700, 300)$.

## Application

**C4.1.** By hand, compute $x_2 = Ax_1$. Then check in software.

**Expected:** $x_2 = (550, 450)$.

**C4.2.** Starting from the same $x_0$, iterate 20 steps (store or print $x_5$ and $x_{20}$). Roughly where do the two populations settle?

**Expected:** toward equal share around $(500, 500)$ (about half on campus).

**C4.3.** Restart from $x_0 = (0, 1000)$. After 20 steps, what happens? In one sentence, what is surprising if you only look at the starting point?

**Expected:** again near $(500, 500)$. Different starts can head toward the same long-run mix — Lab 5 will name the missing idea.

## One-pager

Preview the Lab 4 ticket; write on the hard copy in class.